

Email History API Fix - Documentation

Issue Summary

The `/api/admin/emails/history` endpoint was showing errors related to the `EmailNotification.sponsorApplicationId` column. The error message suggested the column didn't exist in the database, causing the admin dashboard to display "failing" status for email functionality.

Root Cause Analysis

After investigation, the root cause was identified as:

1. **Prisma Client Out of Sync:** The Prisma client was not regenerated after the schema was updated to include the `sponsorApplicationId` field.
2. **Database Schema:** The database actually HAD the column, but the Prisma client was using an outdated generated schema.
3. **No Error in Schema:** The Prisma schema correctly defined the field with proper relations to `SponsorApplication`.

Schema Definition (Confirmed Correct)

```
model EmailNotification {  
  // ... other fields ...  
  sponsorApplicationId String?  
  // ... other fields ...  
  sponsorApplication SponsorApplication? @relation(fields: [sponsorApplicationId], references: [id], onDelete: Cascade)  
  
  @@index([sponsorApplicationId])  
}
```

Solution Implemented

1. Prisma Client Regeneration

- Ran `npx prisma generate` to regenerate the Prisma client
- This ensured the client matched the current schema definition

2. Database Verification

- Created migration script: `scripts/migrate-email-schema.js`
- Verified that the `sponsorApplicationId` column exists
- Confirmed foreign key constraint is in place
- Verified index on `sponsorApplicationId` exists

3. Testing

- Created test script: `scripts/test-email-history-endpoint.js`
- Verified all query operations work correctly:
- Basic email queries with user relations

- Filtering by `sponsorApplicationId`
- `GroupBy` statistics queries

Files Created

1. `scripts/migrate-email-schema.js` (and `.ts` version)
 - Migration script to add `sponsorApplicationId` column if missing
 - Adds foreign key constraint and index
 - Safe to run multiple times (checks before adding)
2. `scripts/test-email-history-endpoint.js`
 - Test script to verify endpoint functionality
 - Simulates the actual queries used by the API endpoint
 - Useful for debugging future schema issues
3. `scripts/check-email-schema.ts`
 - Diagnostic script to inspect `EmailNotification` table structure
 - Queries database schema information
4. `scripts/add-sponsor-application-column.sql`
 - SQL migration script
 - Can be run directly on the database if needed

Testing Results

All tests passed successfully:

-  Column exists in database
-  Foreign key constraint present
-  Index on `sponsorApplicationId` present
-  Prisma queries work correctly
-  Email history endpoint queries functional

Prevention Measures

To prevent similar issues in the future:

1. Always Run Prisma Generate After Schema Changes

```
npx prisma generate
```

2. Use Postinstall Hook

The project already has a `postinstall` script that runs `prisma generate`. Ensure it's working:

```
"scripts": {  
  "postinstall": "prisma generate"  
}
```

3. Deployment Checklist

When deploying schema changes:

- [] Update `schema.prisma`

- [] Run `npx prisma generate` locally
- [] Test queries locally
- [] Run `prisma db push` or create migration
- [] Verify on staging/production
- [] Regenerate Prisma client on deployment

4. CI/CD Integration

Ensure your build process includes:

```
prisma generate && next build
```

(Already present in the project's build script)

API Endpoint Status

The `/api/admin/emails/history` endpoint now works correctly with:

- Pagination support
- Filtering by status, type, userId
- Search functionality
- Date range filtering
- Statistics aggregation
- Type distribution
- Proper inclusion of user and videoAnalysis relations
- Support for sponsorApplicationId foreign key

Usage Example

API Request

```
GET /api/admin/emails/history?page=1&limit=20&status=SENT
```

Response

```
{
  "emails": [
    {
      "id": "...",
      "recipientEmail": "user@example.com",
      "subject": "Welcome to Mindful Champion",
      "status": "SENT",
      "type": "WELCOME",
      "userId": "...",
      "sponsorApplicationId": null,
      "videoAnalysisId": null,
      "user": { ... },
      "contentPreview": "..."
    }
  ],
  "pagination": { ... },
  "statistics": { ... },
  "typeDistribution": [ ... ]
}
```

Verification Commands

Check Column Exists

```
node scripts/migrate-email-schema.js
```

Test Endpoint Queries

```
node scripts/test-email-history-endpoint.js
```

Regenerate Prisma Client

```
npx prisma generate
```

Conclusion

The issue was resolved by regenerating the Prisma client to match the current schema. The database was already correctly configured. The email sending functionality remains intact, and the admin dashboard should now correctly display email history without errors.

Date Fixed: December 14, 2025

Fixed By: DeepAgent

Related Issue: Admin dashboard showing “failing” for email history endpoint